

# Resolvent-controlled pure-dyadic recursive-chain uniform neighbourhood gap

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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## Abstract

R-161 closes a cardinality-uniform neighbourhood only for one-shot raw-source affine controls. This note instantiates an actual predictable shifted-state law. For every finite acyclic chain  $\eta_0 = \xi_0, \eta_{j+1} = \xi_{j+1} + A_j \eta_j$  on the retained pure-dyadic modes  $p, 2p, \dots, 2^J p$ , it proves one positive analytic coefficient radius, uniform in the finite chain length, nonzero starting mode, finite cutoff, and admitted common-even contraction regulator. Inside that radius the complete expected *controller-pullback* Hessian of the globally summed terminal action is larger than 13/100 in coefficient norm and larger than 1/10 in the induced physical-source tangent metric. The proof includes the projected-force connection and the full sixth-power connection; neither may be discarded after nonlinear pullback. The theorem does not cover unrelated multi-chain forests, random or nonlinear coefficient feedback, revisits, a pathwise fibrewise conditional Hessian, or T-050.

## 1 Purpose and scope: result and exact boundary

Work on the fixed side-16 three-torus, in physical dimension three, with the hash-pinned A1 symbol, the fixed positive A7 floor, exact nonaliased continuum integration, and one common real-even covariance-matched scalar contraction multiplier. Let  $p \neq 0$  be retained and let

$$p_j = 2^j p, \quad j = 0, \dots, J. \quad (1.1)$$

The independent whitened both-sign real Gaussian blocks are denoted by  $\xi_j$ . For deterministic matrices  $A_j$  of the registered block size, define the temporally ordered recursion

$$\eta_0 = \xi_0, \quad h_{j+1} = A_j \eta_j, \quad \eta_{j+1} = \xi_{j+1} + h_{j+1}. \quad (1.2)$$

Thus the later controller reads the actual shifted state  $\eta_j$ , not the raw innovation  $\xi_j$ . Put

$$\|A\|_{\mathfrak{E}}^2 = \sum_{j=0}^{J-1} \|A_j\|_{\text{HS}}^2. \quad (1.3)$$

**R-162.** There exists an explicit analytic number  $\delta_* > 0$ , independent of  $J, p$ , the finite retained cutoff, and the admitted multiplier, such that  $\|A\|_{\mathfrak{E}} < \delta_*$  implies, for every  $H \neq 0$ ,

$$\boxed{D^2(\mathcal{A} \circ \chi)_A[H, H] > \frac{13}{100} \|H\|_{\mathfrak{E}}^2 > \frac{1}{10} \mathcal{G}_A(H, H).} \quad (1.4)$$

Here  $\chi(A) = h(A)$ ,  $\mathcal{G}_A(H, H) = \mathbb{E}\|D\chi_A[H]\|^2$ , and  $\mathcal{A}$  is the one globally summed R-141 production scalar: source cost, complete global endpoint/current owner, and terminal sixth-power term, each included

exactly once. Equation (1.4) is a controller-pullback theorem. No intrinsic affine physical-source Hessian is promoted in R-162.

The radius is analytic and deliberately nonnumerical. Its compact-data derivative inputs are proved finite but are not evaluated numerically. No finite-dimensional family-by-family continuity argument is used.

## 2 The recursive transfer and its exact jets

Let  $\mathfrak{E}_J = \bigoplus_{j=0}^J E_j$  and define the one-step block shift

$$(N_A x)_{j+1} = A_j x_j, \quad (N_A x)_0 = 0. \quad (2.1)$$

It is nilpotent. Set

$$T_A = (\text{Id} - N_A)^{-1}, \quad B_A = T_A - \text{Id} = N_A T_A. \quad (2.2)$$

Then  $\eta = T_A \xi$  and  $h = B_A \xi$ . If  $r = \|A\|_{\mathfrak{E}} < 1$  and  $\tau = (1 - r)^{-1}$ , then

$$\|N_A\|_{\text{op}} \leq \|N_A\|_{\text{HS}} = r, \quad \|T_A\|_{\text{op}} \leq \tau, \quad \|B_A\|_{\text{HS}} \leq r\tau. \quad (2.3)$$

These estimates contain no factor depending on the path length.

For coefficient directions  $H_1, \dots, H_m$ , exact resolvent differentiation gives

$$D^m T_A[H_1, \dots, H_m] = \sum_{\sigma \in S_m} T_A N_{H_{\sigma(1)}} T_A \cdots N_{H_{\sigma(m)}} T_A. \quad (2.4)$$

Consequently, for  $m = 1, 2, 3$ ,

$$\|D^m T_A[H_1, \dots, H_m]\|_{\text{HS}} \leq m! \tau^{m+1} \prod_{a=1}^m \|H_a\|_{\mathfrak{E}}. \quad (2.5)$$

The same derivatives apply to  $B_A$ . In particular,

$$\|D\chi_A[H]\|_{L_\xi^2} \leq \tau^2 \|H\|_{\mathfrak{E}}, \quad \|D^2\chi_A[H, K]\|_{L_\xi^2} \leq 2\tau^3 \|H\|_{\mathfrak{E}} \|K\|_{\mathfrak{E}}. \quad (2.6)$$

Since  $N_H = T_A^{-1}(DB_A[H])T_A^{-1}$ , one also has

$$(1 + r)^{-4} \|H\|_{\mathfrak{E}}^2 \leq \mathcal{G}_A(H, H) \leq \tau^4 \|H\|_{\mathfrak{E}}^2. \quad (2.7)$$

The complete source cost is not the affine surrogate. It is

$$\mathcal{A}_{\text{CM}}(A) = \frac{9}{20} \|B_A\|_{\text{HS}}^2. \quad (2.8)$$

For  $r \leq 1/2$ , the exact product rule and (2.5) yield

$$\|D^3 \mathcal{A}_{\text{CM}}(A)\|_{\text{op}} \leq \frac{27}{5} (1 + r) \tau^5 \leq \frac{1296}{5}. \quad (2.9)$$

Thus the source projected-force connection is paid inside one scalar third-derivative bound.

### 3 Origin transfer: the acceleration must cancel

At  $A = 0$ , the tangent is the R-160 one-shot family:

$$D\chi_0[H] = N_H \xi. \quad (3.1)$$

The recursive chart nevertheless has nonzero acceleration,

$$D^2\chi_0[H, K] = (N_H N_K + N_K N_H) \xi. \quad (3.2)$$

Only adjacent coefficient blocks survive. In the output block  $j + 1$ ,

$$(D^2\chi_0[H, K])_{j+1} = (H_j K_{j-1} + K_j H_{j-1}) \xi_{j-1}. \quad (3.3)$$

It follows directly that

$$\|D^2\chi_0[H, K]\|_{L^2_\xi} \leq 2\|H\|_{\mathfrak{E}}\|K\|_{\mathfrak{E}}, \quad (3.4)$$

uniformly in  $J$ .

Every term in (3.3) is a  $p_j \rightarrow 4p_j$  connection. Against the original carrier it has only Fourier support  $\{\pm 3p_j, \pm 5p_j\}$ . The R-155 centered stationary common-even exact-torus cancellation therefore kills its pairing with the complete global first variation, term by term. The source connection vanishes because  $B_0 = 0$ . This is a global-action statement; it does not identify a local  $E_{\text{CN}}$  atom with  $P_{\text{comp}}$ .

Equivalently, if  $P_A = T_A T_A^*$ , then

$$D^2 P_0[H, H] = 2N_H N_H^* + 2N_H^2 + 2(N_H^*)^2. \quad (3.5)$$

The first term is the R-160 affine covariance Hessian. The last two terms are exactly the cancelled acceleration directions. Hence the recursive pullback at the origin equals the R-160 affine Hessian and obeys

$$\boxed{D^2(\mathcal{A} \circ \chi)_0[H, H] > \frac{4}{25}\|H\|_{\mathfrak{E}}^2.} \quad (3.6)$$

Aliasing, a zero mode, a nonzero mean, mismatched regulators, or a local-owner substitution would invalidate this transfer and are excluded.

### 4 Dyadic synthesis and covariance data

Write  $x_j = |p_j|^2 = 4^j x_0$ . R-161 gives, with

$$c_0 = \frac{3}{20}, \quad g = \frac{244140625000000000}{28800000000947494031}, \quad (4.1)$$

the pointwise value and stacked-current synthesis bounds

$$s_j^2 \leq \frac{g}{1 + x_j^2}, \quad d_j^2 \leq \frac{g x_j}{1 + x_j^2}. \quad (4.2)$$

The single-chain geometry makes both raw sums finite:

$$\begin{aligned} B_S &:= \sum_j s_j^2 \leq \frac{16g}{15c_0^2} = \frac{3125000000000000000}{777600000025582338837}, \\ B_D &:= \sum_j d_j^2 \leq \frac{4g}{3c_0} = \frac{1953125000000000000}{259200000008527446279}. \end{aligned} \quad (4.3)$$

These constants are uniform in  $J, p$ , cutoff, and multiplier.

Let  $S$  and  $D_i$  denote the combined value and current syntheses along the chain. The centered deterministic recursion remains a linear image of the independent Gaussian vector. With

$$R_A = P_A - \text{Id}, \quad (4.4)$$

the exact covariance data after primitive cancellation are

$$\begin{aligned} C_A &= C_0 + SR_AS^*, \\ U_A &:= Q_A - Q_0 = \sum_{i=1}^3 D_i R_A D_i^*, \\ K_{A,i} &= SR_A D_i^*. \end{aligned} \quad (4.5)$$

Common evenness supplies  $K_0 = 0$ . The possibly divergent raw  $Q_0$  is cancelled before any norm is taken. Use the R-161 data norm

$$\|(C, U, K)\|_{\mathcal{Z}}^2 = \|C\|_{\text{op}}^2 + \|U\|_{\text{op}}^2 + \sum_{i=1}^3 \|K_i\|_{\text{HS}}^2. \quad (4.6)$$

For every bounded block operator  $R$ , elementary rank and triangle bounds give

$$\begin{aligned} \|SR_S^*\|_{\text{op}} &\leq B_S \|R\|_{\text{op}}, \\ \left\| \sum_i D_i R D_i^* \right\|_{\text{op}} &\leq 3B_D \|R\|_{\text{op}}, \\ \left( \sum_i \|SR D_i^*\|_{\text{HS}}^2 \right)^{1/2} &\leq (18B_S B_D)^{1/2} \|R\|_{\text{op}}. \end{aligned} \quad (4.7)$$

Thus the explicit finite constant

$$c_Z^2 = B_S^2 + 9B_D^2 + 18B_S B_D < 1 \quad (4.8)$$

controls the linear map in (4.5).

The resolvent product rule gives, for  $m = 1, 2, 3$ ,

$$\|D^m P_A[H_1, \dots, H_m]\|_{\text{op}} \leq (m+1)! \tau^{m+2} \prod_a \|H_a\|_{\mathfrak{E}}. \quad (4.9)$$

Consequently, for  $Y_A = (C_A, U_A, K_A)$ ,

$$\|D^m Y_A[H_1, \dots, H_m]\|_{\mathcal{Z}} \leq c_Z (m+1)! \tau^{m+2} \prod_a \|H_a\|_{\mathfrak{E}}. \quad (4.10)$$

## 5 One global scalar, including the sixth power

Let

$$\overline{\mathcal{B}}(C) = \mathbb{E}_{N(0,C)} \mathcal{B}(W), \quad \overline{\mathcal{T}}(C) = \mathbb{E}_{N(0,C)} D^2 \mathcal{B}(W). \quad (5.1)$$

The globally summed R-107/R-141 endpoint identity and the centered Gaussian sixth moment combine into the single pointwise scalar

$$\begin{aligned} \Phi(C, U, K) &= \frac{1}{2} \left\{ \text{Tr}[U \overline{\mathcal{B}}(C)] + \sum_i K_i \otimes K_i : \overline{\mathcal{T}}(C) \right\} \\ &\quad + \frac{3}{20} \left[ (\text{Tr } C)^3 + 6 \text{Tr } C \text{Tr}(C^2) + 8 \text{Tr}(C^3) \right]. \end{aligned} \quad (5.2)$$

Up to an irrelevant constant, the full expected action is exactly

$$F(A) = \frac{9}{20} \|B_A\|_{\text{HS}}^2 + \int_{\mathbb{T}_{16}^3} \Phi(Y_A(x)) dx. \quad (5.3)$$

This placement of the sixth power is load-bearing. Its intrinsic affine physical-source Hessian is positive semidefinite, but its nonlinear controller pullback contains the indefinite projected-force/acceleration term. For the scalar test

$$f(z) = z^6, \quad z(a, b) = ab, \quad (5.4)$$

the Hessian of  $a^6 b^6$  at  $(1, 1)$  is

$$\begin{pmatrix} 30 & 36 \\ 36 & 30 \end{pmatrix}, \quad (5.5)$$

with eigenvalues 66 and  $-6$ . Therefore R-161's affine step of dropping the global positive sextic Hessian cannot be reused here.

For  $r \leq 1/2$ ,  $\|P_A - \text{Id}\|_{\text{op}} \leq \tau^2 - 1 \leq 3$ . R-161's background bound and (4.7) place every  $Y_A(x)$  in one compact finite-dimensional covariance-data box, independent of the chain. The positive A7 floor and the Gaussian heat identity extend the derivatives of the first line of (5.2) continuously to singular positive-semidefinite  $C$ ; the third derivative uses at most  $D^8 \mathcal{B}$ . The second line is an explicit covariance polynomial. Hence

$$M_j = \sup_{Y \text{ in the box}} \|D^j \Phi(Y)\|_{\text{op}} < \infty, \quad j = 1, 2, 3, \quad (5.6)$$

uniformly in  $J, p$ , cutoff, and multiplier.

## 6 Uniform Hessian modulus and retained gap

At  $r_0 = 1/2$ , one has  $\tau_0 = 2$ , and (4.10) gives the three constants

$$J_1 = 16c_Z, \quad J_2 = 96c_Z, \quad J_3 = 768c_Z. \quad (6.1)$$

The complete third-order chain rule is

$$\|D^3(\Phi \circ Y)\| \leq M_3 J_1^3 + 3M_2 J_1 J_2 + M_1 J_3. \quad (6.2)$$

The  $M_1 J_3$  term is mandatory: recursive covariance has a nonzero third jet. With torus volume  $16^3 = 4096$ , define

$$L_* = \frac{1296}{5} + 4096(4096M_3 c_Z^3 + 4608M_2 c_Z^2 + 768M_1 c_Z). \quad (6.3)$$

Equations (2.9), (4.10), and (6.2) prove

$$\|D^2 F(A) - D^2 F(0)\|_{\text{op}} \leq L_* \|A\|_{\mathfrak{E}}, \quad \|A\|_{\mathfrak{E}} \leq \frac{1}{2}. \quad (6.4)$$

Every forward block and its legal reverse are adjoints inside this one self-adjoint Hessian. Their balanced term is the symmetric cross, not a third payment. The derived states  $\eta_j$  are not independent low or Feshbach variables.

Choose

$$\delta_* = \min \left\{ \frac{1}{2}, \frac{3}{100(1 + L_*)} \right\} > 0. \quad (6.5)$$

Then  $L_* \delta_* < 3/100$ , and (3.6) gives

$$D^2 F(A)[H, H] > \left( \frac{4}{25} - \frac{3}{100} \right) \|H\|_{\mathfrak{E}}^2 = \frac{13}{100} \|H\|_{\mathfrak{E}}^2. \quad (6.6)$$

Moreover  $\delta_* < 3/100$ , while

$$\left(\frac{100}{97}\right)^4 < \frac{13}{10}. \quad (6.7)$$

Using (2.7), (6.6) therefore implies

$$D^2F(A)[H, H] > \frac{1}{10}\mathcal{G}_A(H, H), \quad (6.8)$$

which completes (1.4).

The pullback identity represented by (5.3)–(6.8) is precisely

$$D^2(\mathcal{A} \circ \chi)_A[H, K] = \langle D\chi_A H, H_{\text{prod}}(h_A) D\chi_A K \rangle + \langle g_{\text{prod}}(h_A), D^2\chi_A[H, K] \rangle. \quad (6.9)$$

The second term contains the source, endpoint/current, trace, and sextic first variations once. No owner is added a second time.

## 7 What failed and what remains open

1. **Dropping the sextic as positive failed.** The exact fixture (5.4)–(5.5) has a negative pullback eigenvalue. Repair: include the full Gaussian sextic polynomial in  $\Phi$  and retain all three terms in (6.2).
2. **Bare finite-dimensional continuity failed as a uniform proof.** The registered coherent fixture

$$\mathcal{F}_N(h) = \frac{2}{25}\|h\|_2^2 - \frac{(\sum_j h_j)^4}{4N} + \frac{(\sum_j h_j)^6}{6N^2} \quad (7.1)$$

has origin Hessian  $4/25$ , but along  $h = N^{-1}(1, \dots, 1)$  its coherent Hessian is  $4/25 - 3 + 5/N < 0$  for  $N \geq 3$ . Repair: use the explicit chain-length-free resolvent and dyadic synthesis bounds.

3. **A stationarity-free abstract resolvent transfer failed.** A positive intrinsic quadratic endpoint can still have a negative recursive pullback if its origin gradient pairs with the  $p \rightarrow 4p$  acceleration. R-162 survives only because Section 3 proves the exact full-action Fourier cancellation. The abstract counterexample is not an A1 counterexample.
4. **The single-chain raw-current bound does not union over an arbitrary forest.** Across all unrelated three-dimensional roots the crude sum of  $d_p^2 \sim |p|^{-2}$  grows with the cutoff. A multi-chain result needs a new full-lattice weighted recursive estimate, not repetition of (4.3).
5. **Random nonlinear coefficients leave the Gaussian chart.** Already  $A(\xi) = \varepsilon \operatorname{sgn} \xi$  sends a centered Gaussian to a component containing  $\varepsilon|\xi|$ , which is noncentered and non-Gaussian. The covariance scalar (5.2) no longer determines the law.

R-162 therefore advances from the fixed two-stage R-159 chart to every finite acyclic pure-dyadic deterministic-matrix shifted-state chain, uniformly in chain length. It does not prove an intrinsic physical-source Hessian, multiple unrelated chains, random or nonlinear past-dependent coefficients, revisits or cycles, a pathwise fibrewise conditional inequality, cutoff or floor removal, T-050, A13, Nelson, an interacting measure, any phase or PDE, or Sector-A closure.

## 8 Devil’s-advocate audit

- **Objection:** the R-160 origin theorem is affine, so recursion adds an unpaid second jet. **VALID WITH MITIGATION.** Equations (3.2)–(3.5) expose that jet. Its complete global first-variation pairing vanishes only by the registered stationary exact-Fourier cancellation.
- **Objection:** convexity of  $|W|^6$  makes the sextic harmless. **UPHELD AGAINST THAT SHORTCUT.** Convexity controls intrinsic affine directions, not nonlinear pullback curvature. The  $-6$  eigenvalue in (5.5) forces the sextic into the common modulus.
- **Objection:** the divergent raw derivative covariance was hidden inside a norm. **DISMISSED.** Equation (4.5) forms  $Q - Q_0$  before norms. Only the dyadic-chain synthesis  $B_D$  appears.
- **Objection:** path length has leaked into a matrix dimension or compactness constant. **DISMISSED.** The resolvent bounds use  $\|N_A\|_{\text{op}} \leq \|A\|_{\ell^2(\text{HS})}$ , and the data box uses only the geometric sums (4.3) and fixed six-real output rank.
- **Objection:** (1.4) proves the general T-050 conditional Hessian. **UPHELD AS PROHIBITED.** R-162 is expected/global, single-chain, centered Gaussian, deterministic-matrix, acyclic, fixed-floor, and finite cutoff. The missing cases listed after Section 7 remain substantive gates.

## 9 Reproduction and Result footer

Primary command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_resolvent_pure_dyadic_recursive_chain_uniform_  
neighborhood_gap_boundary.py
```

Independent command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_resolvent_pure_dyadic_recursive_chain_uniform_  
neighborhood_gap_boundary_independent.py
```

Integrated command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_resolvent_pure_dyadic_recursive_chain_uniform_  
neighborhood_gap_boundary_verify.py
```

Result ID: A13-CLASSII-RESOLVENT-PURE-DYADIC-RECURSIVE-CHAIN-  
UNIFORM-NEIGHBORHOOD-GAP-BOUNDARY; Ledger ID: R-162.

Dependencies: A1, A7, R-107, R-141, R-155, R-159, R-160, R-161.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe  
codes/foundations/  
a13\_classii\_resolvent\_pure\_dyadic\_recursive\_chain\_uniform\_  
neighborhood\_gap\_boundary\_verify.py

Precise statement: one positive analytic  $\ell^2(\text{HS})$  radius, uniform in finite  
chain length, mode, cutoff and regulator, retains pullback gap  $>13/100 > 1/10$ .

Scope: side-16 d=3 A1/A7, exact torus, centered independent Gaussian blocks,  
common-even matched multiplier, deterministic acyclic shifted-state chain.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary, independent, integrated, PDF and release PASS.

Falsification gate: any resolvent, origin, synthesis, connection, gap, hash,  
PDF, public-surface, exploration, or release assertion fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no intrinsic/forest/random/revisit/pathwise-conditional,  
removal, T-050/A13, Nelson, measure, phase/PDE, or Sector-A theorem.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: test an actual forest or truth-first surviving law.